← Back **Week 5 quiz**
Graded Assignment • 20 min

Due Mar 16, 11:59 PM CET

1. What causes a meteor shower? 1 point

- ○ The Earth passes nearby to a comet, which grazes the atmosphere and bounces off.
- ○ The Earth passes nearby to a comet, and the comet strikes the Earth's surface.
- ◉ The Earth's orbit passes through a stream of debris left by a comet, and most of this debris burns up in the atmosphere.
- ○ The Earth passes nearby to a comet, which breaks up into large chunks on entering the Earth's atmosphere.

2. If a CCD pixel can hold sixteen bits per pixel, and the CCD chip is a square with 1000 pixels on a side, approximately how many bytes can the CCD hold? (Note: we are using the "decimal" definition of megabytes, gigabytes etc, so one kilobyte = 1000 bytes, one megabyte = 1000 kilobytes and so on). 1 point

- ○ 4 MB
- ○ 1 MB
- ○ 2 GB
- ◉ 2 MB

3. A modern sky survey will contain around one petabyte of data. How many bytes is this? (Remember we are using the decimal representation of bytes, so that one kilobyte = 1000 bytes, and so on). 1 point

- ○ 10^9 bytes
- ○ 10^{18} bytes
- ○ 10^6 bytes
- ◉ 10^{15} bytes
- ○ 10^{12} bytes

4. An astronomer is searching through a catalogue of half a billion objects to identify a small number of targets. She can store the catalogue on her laptop's hard drive. Once the data is loaded from the laptop's hard drive in to memory, the laptop can process the data to find the targets in less than a second. But, each row of the catalogue has a transfer overhead from the hard drive to the memory of 2 milliseconds. She is looking to complete the processing work in 7 days, as she has a deadline to meet. Should she do a brute force search through all the objects? 1 point

- ○ No, because brute force searches are unlikely to find the correct targets.
- ○ No, because the catalogue cannot be read into the CPU completely, and will be read in in chunks.
- ◉ No, because the total operation will take longer than 7 days to process on her laptop, and she will miss her deadline.
- ○ Yes, because a brute force search is the only way to guarantee that she will find all her targets.

5. In video 4, Catherine described an example of tracking an asteroid as it moves on the sky. Over a 7 day period, it moves an angular distance of half a degree. If we assume the asteroid is orbiting the Sun in a circle, then what is the asteroid's orbital period, and how far away is it from the Sun? (The gravitational constant $G = 6.67 \times 10^{-11}\, m^3 s^{-2} kg^{-1}$, the Sun's mass is $M = 1.99 \times 10^{30}\, kg$, and an astronomical unit is: $1AU = 1.496 \times 10^{11} m$. There are $3.16 \times 10^7 s$ in a year. Note, the motion of the Earth should be neglected as well.) 1 point

- ○ The orbital period is 828 years, and the asteroid is 88.2 AU away from the Sun.
- ○ The orbital period is 7100 years, and the asteroid is 369.5 AU away from the Sun.
- ◉ The orbital period is 14 years, and the asteroid is 5.7 AU away from the Sun.
- ○ The orbital period is 14 years, and the asteroid is 14175 AU away from the Sun.